

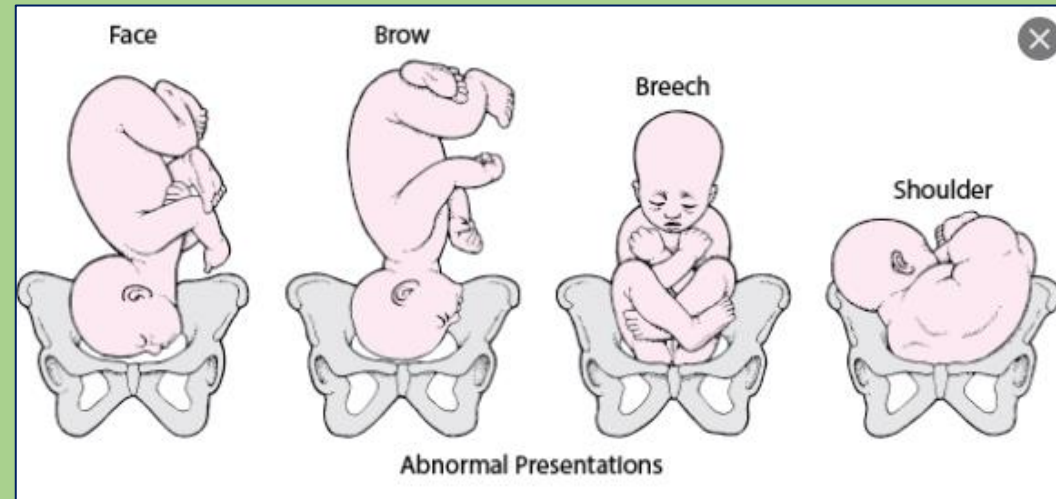
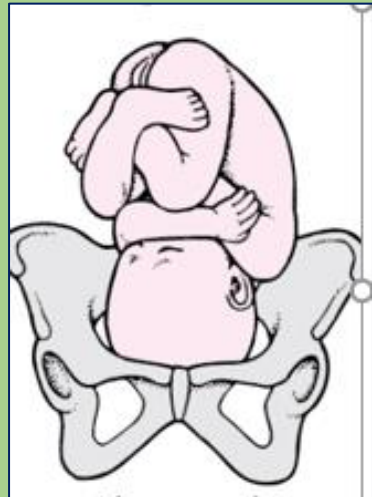
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Malposition and malpresentation in labour

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Learning objectives:

- At the end of this lecture the student should be able to:
- Define normal and abnormal presentation.
- Understand the mechanism of normal labour. mechanism of normal labour
- Define malposition and how to manage labour with malposition.
- Define face, brow and compound presentations.
- How to manage labour with these malpresentation.

- **Mechanism of normal labour:**

- Refers to **the series of changes** in **position and attitude** that the fetus undergoes during its passage through the birth canal.

- Although labour is a continuous process mechanism of labour described in 8 movements:

- **1.Engagement.**

- **2.Descent.**

- **3. Flexion.**

- **4. Internal rotation.**

- **5.Extension.**

- **6. Restitution.**

- **7.External rotation.**

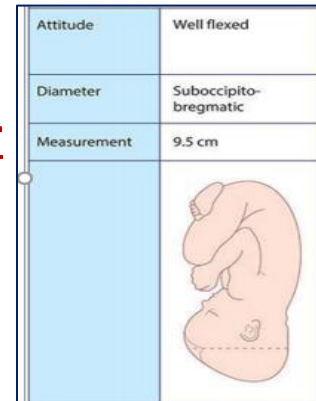
- **8. Delivery of shoulders and body.**

- In normal labour, the fetus enter the pelvis with the occiput in the transverse diameter (occipito-transverse) or in one of the oblique diameters and directed anteriorly (right or left occipito-anterior position).

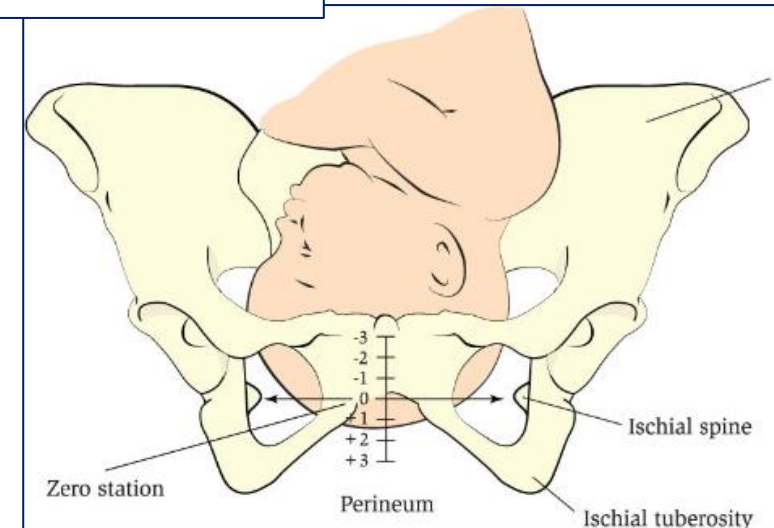
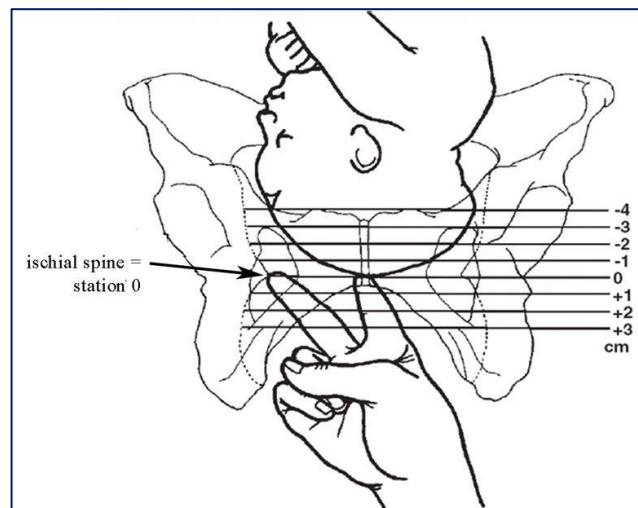
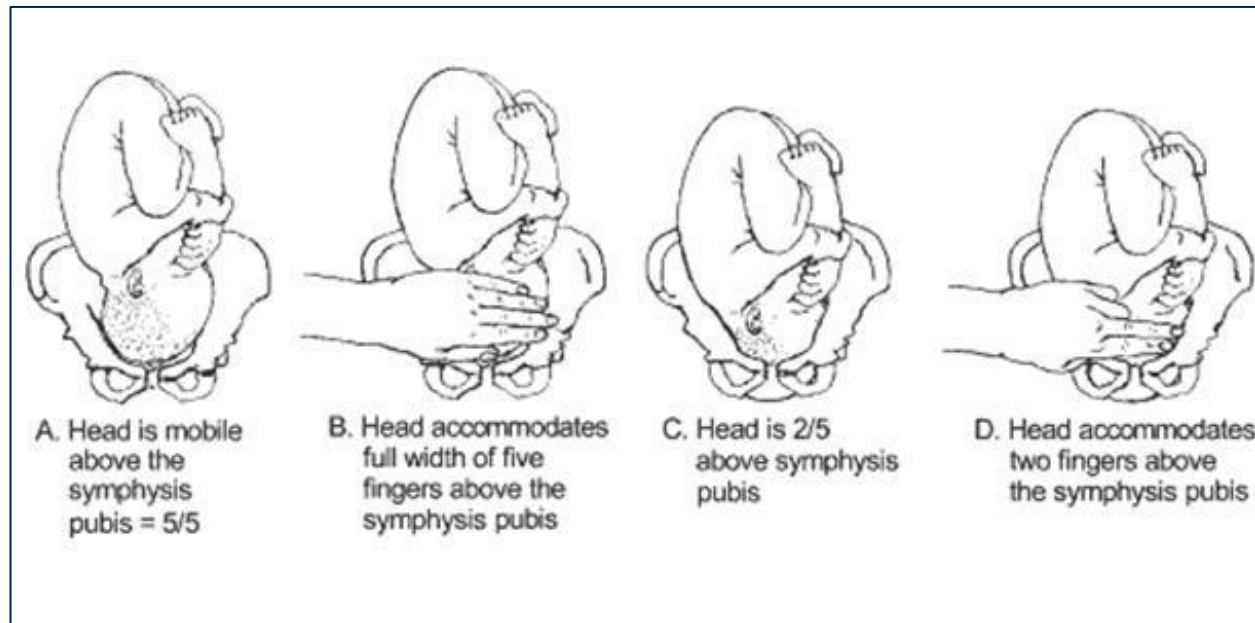
-When the head is well flexed the smallest antero-posterior diameter (the sub-occipito-bregmatic= 9.5cm) pass through the pelvis.

1.Engagement: is said to have occurred when the widest transverse diameter of the presenting part has passed successfully through the pelvic inlet.

- In normal labour with vertex presentation the widest transverse diameter is the Biparietal diameter (9.5cm).
- Engagement occur in the vast majority of nulliparous women prior to labour, usually by 37 weeks' gestation, but not so for the majority of multiparous women (which happened in labour).
- The number of fifths of the fetal head palpable abdominally above the symphysis pubis is used to describe whether engagement has taken place.
- If only two-fifths of the fetal head is palpable abdominally, the head is engaged, if more than 2/5 it is not.



How to assess engagement of the fetal head by abdominal and vaginal examination

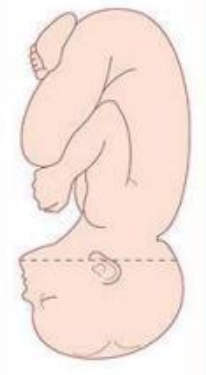


2. Descent: It is a continuous process throughout labour.

3. Flexion: As the head descends into the narrower mid-pelvis, flexion occurs.

-Flexion is very important, it brings the smallest diameter of the fetal head to pass through the pelvis

-The more flexion the smaller diameter.

	Flexed  Extended			
Attitude	Well flexed	Less well flexed (partially extended) or deflexed	Extended 'brow presentation'	Hyperextended 'face presentation'
Diameter	Suboccipito-bregmatic	Occipito-frontal	Occipito-mental	Submento-bregmatic
Measurement	9.5 cm	11.5 cm	13.0 cm	9.5 cm
				

4. Internal rotation:

The fetal head typically enters the pelvis orientated in a transverse position in keeping with the wider transverse diameter of the inlet.

- But at the outlet the A-P diameter is longer than transverse , and the fetal head must rotate from a transverse to an OA position as it passes through the mid-pelvis to enter the pelvic outlet.

- Typically, this happens in the mid-pelvis where the transverse and A-P diameters are similar(12cm).

- When the head is well flexed, the occiput will be at lower level and reach and hit the pelvic floor muscles first.

- These muscle has gutter shape and when contract it will push the occiput forward to lie under the symphysis pubis (direct occipito-anterior position), and the head enter the pelvic outlet and becomes ready for delivery.

5. Extension: After internal rotation, the occiput lies under the symphysis pubis & the head which is fully flexed should undergoes extension in order to be delivered & the sub-occipito-frontal diameter (10cm) will pass through the vulval outlet

6. Restitution: when the head undergoes internal rotation, there will be twisting of the head on the body, after delivery the head undergoes slight lateral rotation toward maternal thigh to correct this twisting.

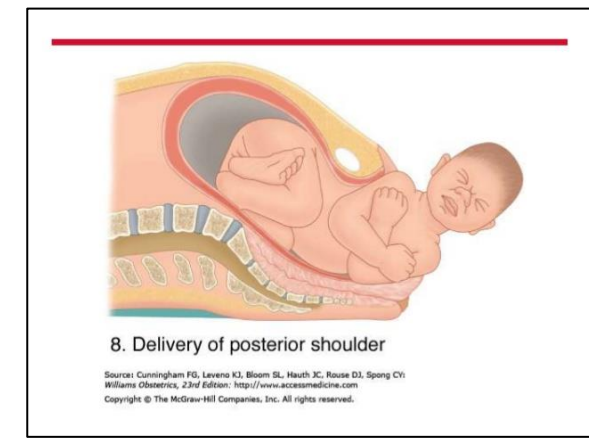
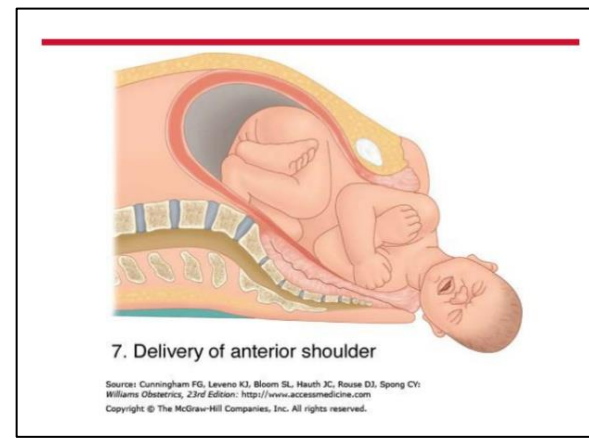
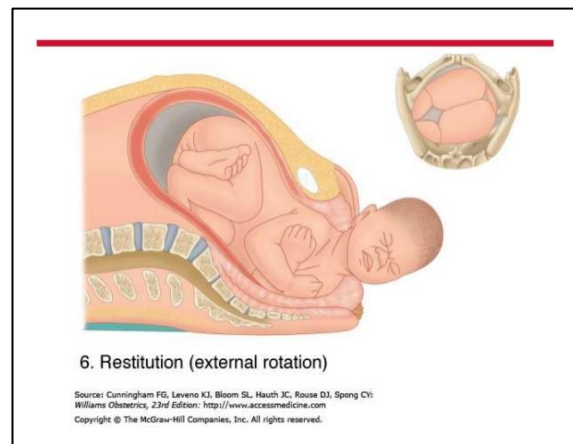
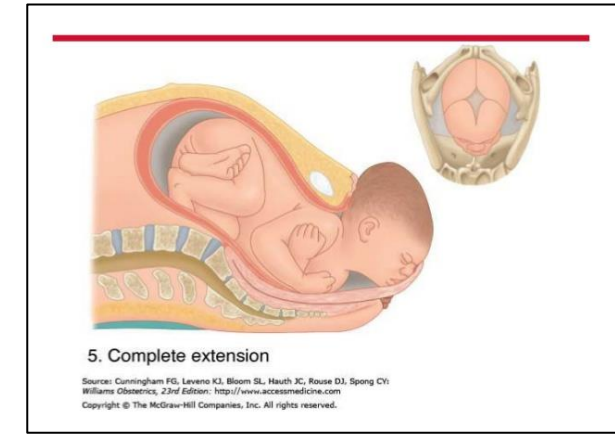
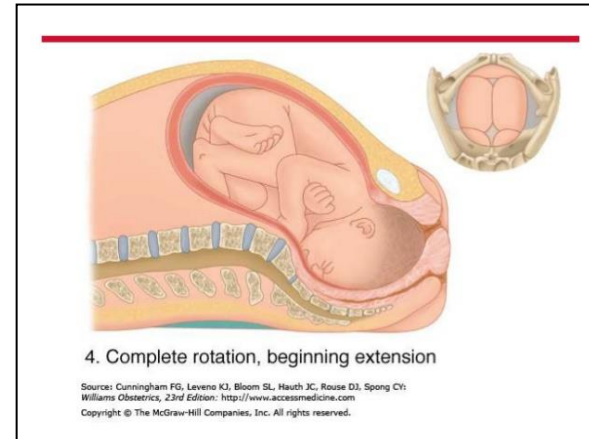
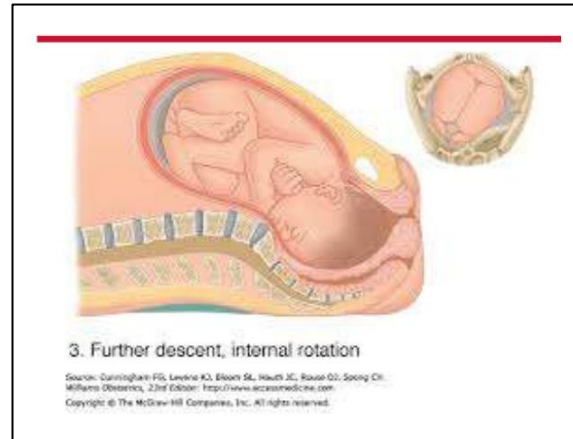
7. External rotation: after delivery of the head, the shoulders enter the pelvis in the transverse diameter then undergoes internal rotation to enter the outlet in AP diameter.

with internal rotation of shoulders inside the pelvis, the head will rotate further toward the maternal thigh externally.

-So External rotation of the head indicate internal rotation of the shoulders.

8. Then the anterior shoulder delivered by downward traction followed by posterior shoulder by upward traction, then the body delivered easily.

Mechanism of normal labour



Malposition of the fetal head :



This term is used when the head enters the pelvis with the occiput directed posteriorly toward one of the sacro-iliac joints either left or right .

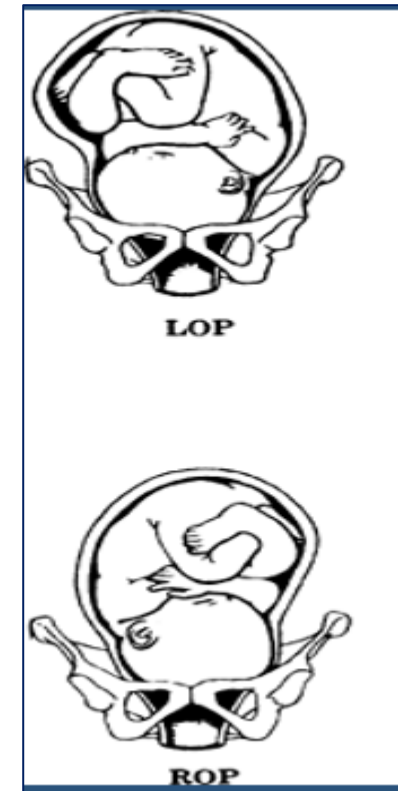
-There are two occipito-posterior position:

- Left occipito-posterior position (LOP):

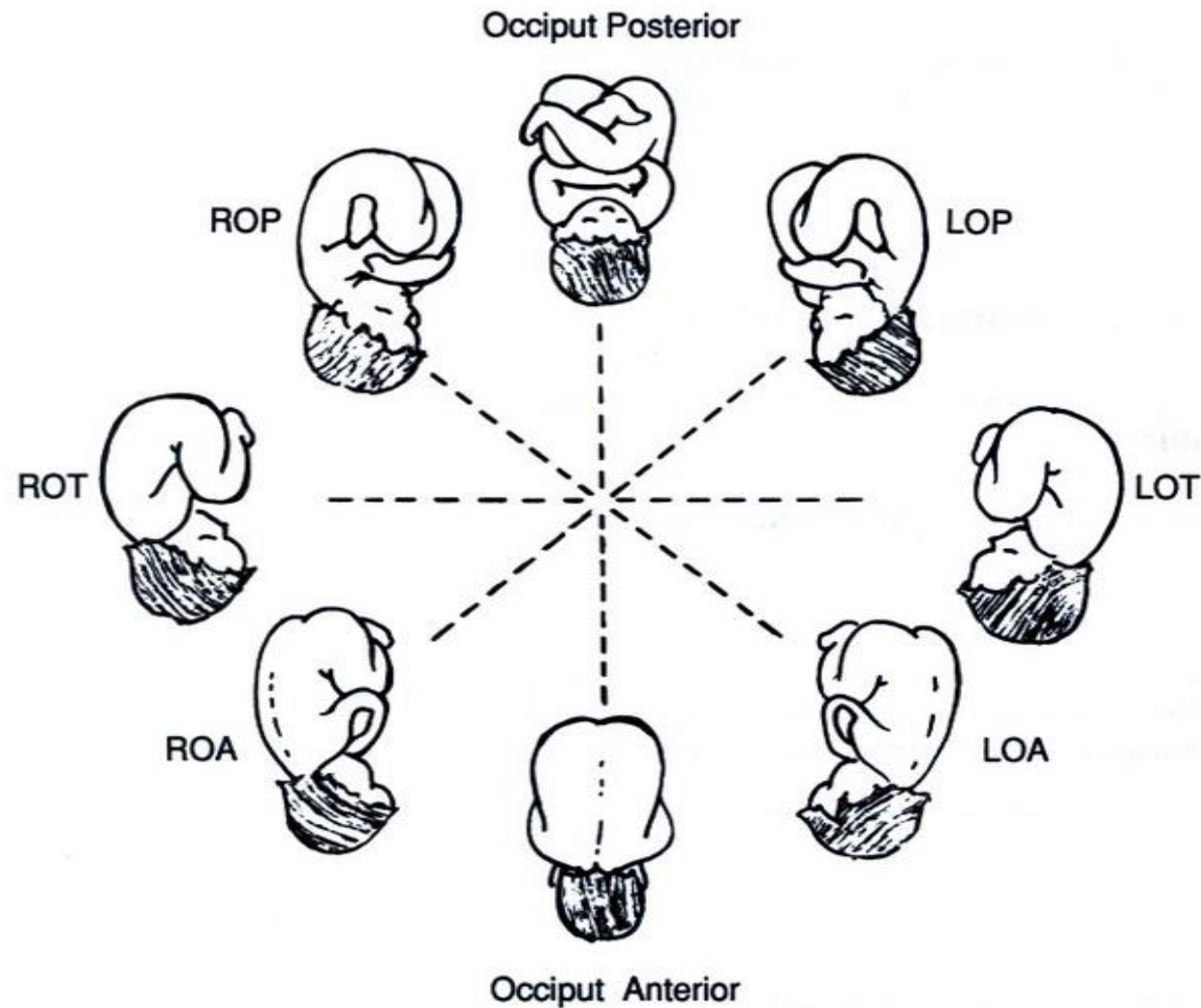
the occiput directed to the left sacro-iliac joint.

- Right occipito-posterior position (ROP):

the occiput directed to the right sacro-iliac joint.



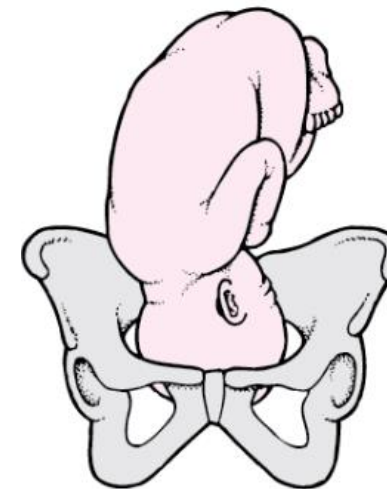
Fetal positions



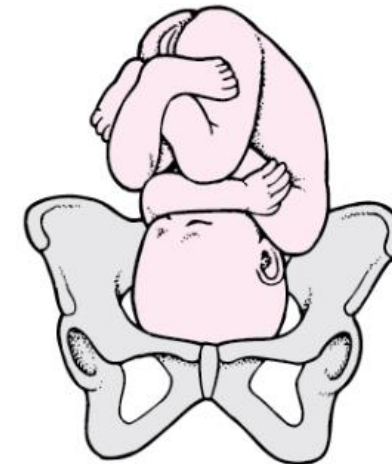
- **Mal-positions** are characterized by minor degrees of de-flexion of the head

and thus present a larger antero-posterior diameter (occipito- frontal) of 11.5 cm to pass through the pelvic inlet.

- Malposition (or occipito-posterior position) of the fetal head is important because it is associated with **labour abnormalities** (**prolonged labour**) that may lead to adverse maternal and neonatal consequences.



Normal Position and Presentation



Abnormal Position

Causes of malposition (occipito-posterior position):

1. Pelvic shape abnormalities e.g.: android pelvis.
2. Anterior placenta.
3. Idiopathic.

The mechanism and outcome of labour:

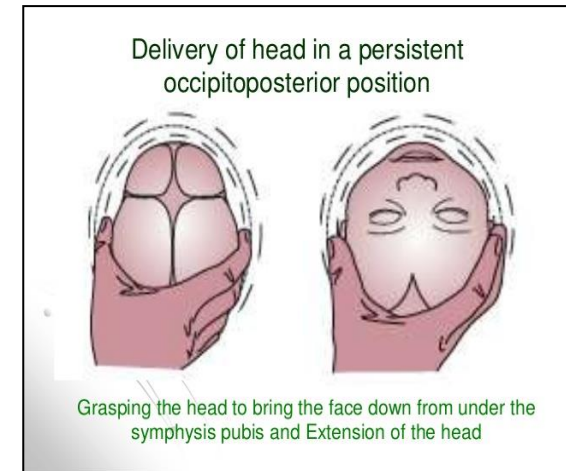
*This will depend on the degree of flexion of fetal and the are three possibilities:

1. When the head is well flexed →

- *The head engaged easily and descend down and reach the pelvic floor.
- *The Occiput will hit the levator ani muscles which will contract and push the occiput forward.
- *So the occiput undergoes long forward rotation from OP to OA position and delivered without problem.

- 2. When the head is incompletely flexed

- The diameter which pass through the pelvis is occipito-frontal diameter which measure 11.5 cm instead of the sub-occipito-bregmatic diameter which measure 9.5 cm (there will be delayed engagement).
- With **incomplete flexion** the sinciput will meet the pelvic floor first and rotate anteriorly to lie behind the symphysis.
- While the occiput rotate backward by one-eighth of the circle to lie in the hollow of the sacrum(lead to direct occipito-posterior position).
- The head may now be born with the face towards the posterior surface of the symphysis pubis (face to pubis position).
- The vertex is born by flexion, followed by the occiput, Then the head extends, so the face and chin emerging from under the pubic arch and delivered.
- The vulval orifice is stretched by the occipito-frontal (11.5cm) instead of the sub-occipito-frontal (10cm) diameter with a difference in size of 1.5 cm. and a severe perineal tears may result.



3.Deep transverse arrest:

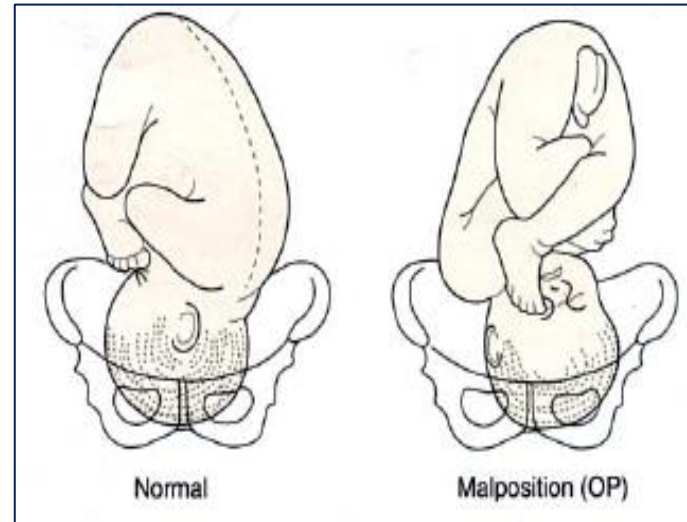
- In some cases the head becomes **arrested with it's long axis in the transverse diameter of the pelvis.**
- The degree of extension being such, that both the occiput and sinciput being at same level
- That the head start forward rotation then the rotation will be arrested with the occiput in the transverse diameter.**
- The diagnosis usually made by vaginal examination during the 2nd stage where the head found to be arrested at the level of the ischial spine with the sagittal suture in the transverse diameter.
- In such condition labour will be arrested and assistance needed to deliver the baby by either instruments or C/S.**

Diagnosis:

During labour:

There is delay engagement of head.

- The position can be determined from the direction of the anterior fontanelle, which can be easily felt behind the symphysis pubis.



The degree of flexion of the head can be determined from the fontanelles also.



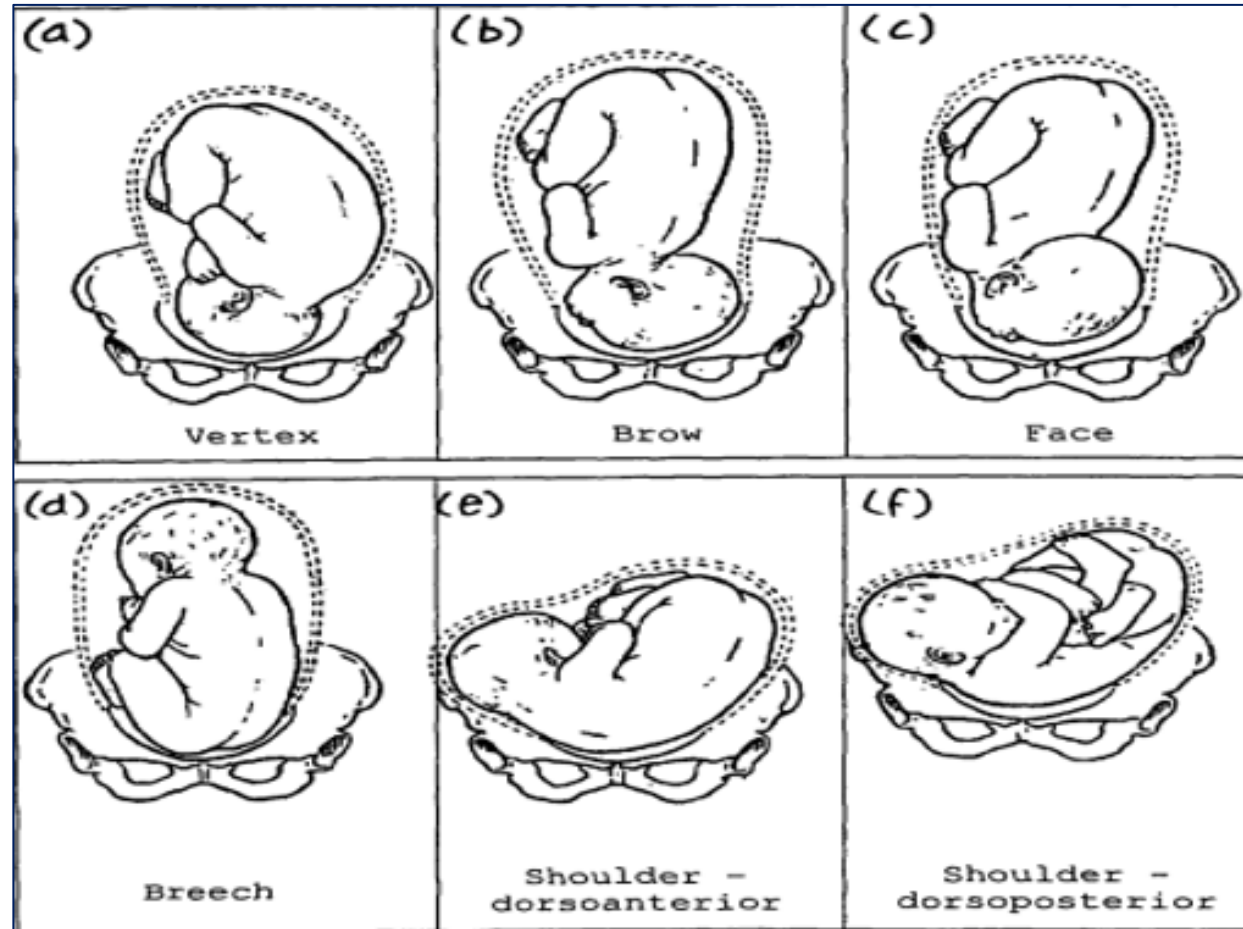
- 1.If only the anterior fontanelle can be felt the head is poorly flexed (deflexed).
 - 2.If both the anterior and posterior fontanelles can be felt the head is less poorly flexed.
 - 3.If only the posterior fontanelle felt the head is well flexed.
- Remember a well flexed head is more likely to rotate anteriorly.

The course of labour in occipito-posterior position:

- Prolongation of the 1st and 2nd stage of labour are common.
- Ineffective uterine contraction is common because the poorly flexed head fails to press down upon the cervix.
- In 80% of cases (head well flexed): there will be spontaneous rotation of the occiput to the anterior position and delivered as OA.
- In about 10% (the head is deflexed): the occiput undergoes short backward rotation and delivered in direct occipito-posterior position (face-to-pubes) with risks of perineal tear.
- In about 10% of cases: there is failure of the presenting part to complete internal rotation and rotation arrested in occipito-transverse position, so the head fail to inter the outlet (Deep Transverse arrest (such baby need delivery by C/S or instruments).
- labour is managed as in OA position, if there is delay in progress augmentation with oxytocin may help rotation which usually happened in the 2nd stage.
- If still no progress the baby should be delivered by C/S.

Malpresentation:

- Any presentation other than vertex presentation is called malpresentation.

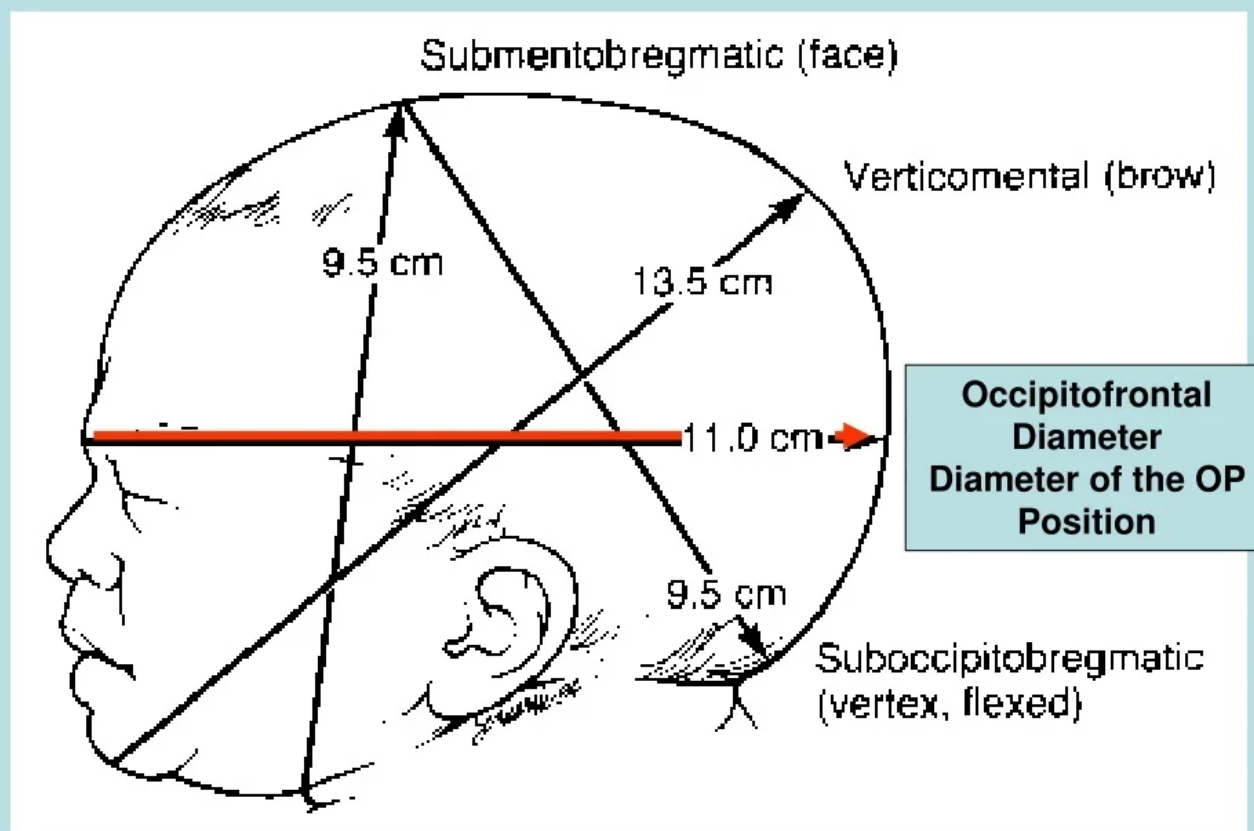


Causes of malpresentation:



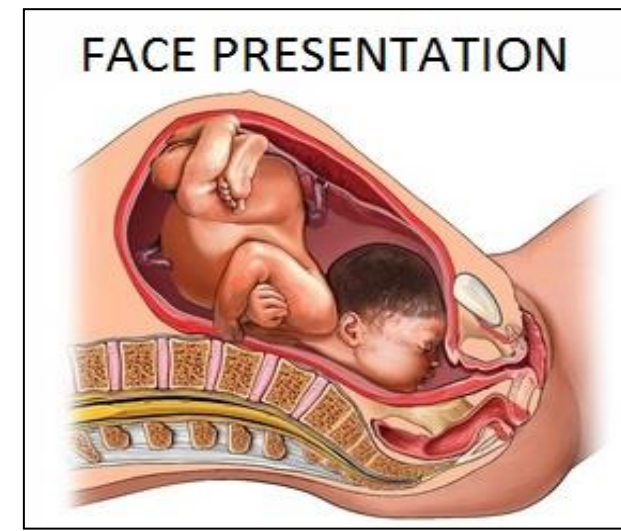
1. Prematurity.
2. Multiple pregnancy.
3. Abnormal baby (anencephaly, hydrocephaly .
4. Polyhydramnios or Oligohydramnios.
5. Congenital malformation of the uterus(uterine septum or bi-cornuate uterus.
6. Placenta praevia.
7. Contracted pelvis.
8. Mass in the uterus e.g. uterine fibroid.

Fetal presenting diameter according to presentation:



Face presentation):

- In a face presentation, the fetal head and neck are hyper-extended,
- The presenting portion of the fetus is the fetal face between the orbital ridges and the chin.
- Incidence is One in 500 labours (0.2%).



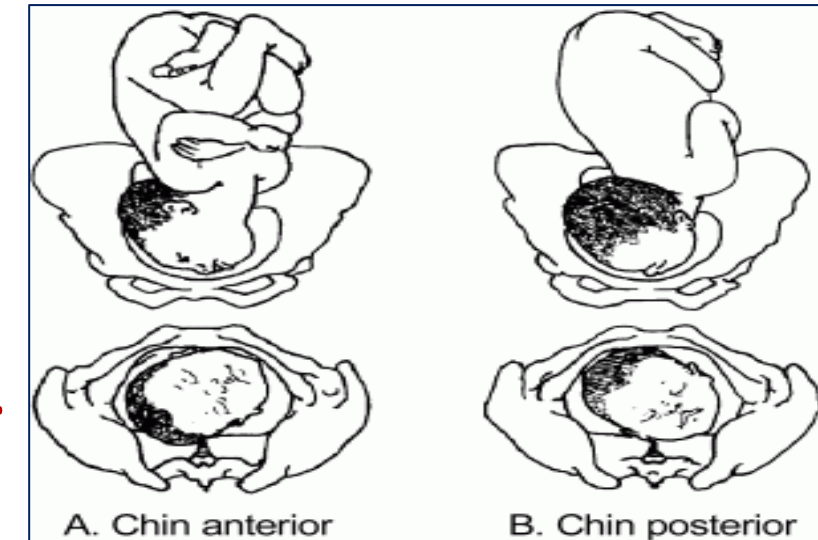
Mechanism of labour: depends on the position:

The fetal chin (mentum) is the part used to define the position during vaginal examination.

So four positions can present:

- Right mento-anterior.
- Left mento-anterior.
- Left mento-posterior (to left sacro-iliac joint).
- Right mento-posterior(to right sacro-iliac joint).

Mento-anterior positions are more common (over 60% of cases).



Mechanism of labour



If the head enter the pelvis In a mento-anterior position:

- The head engages and descend with increasing extension.
- So the sub-mento-bregmatic diameter (9.5cm) comes through the pelvic inlet.
- When the chin reaches the pelvic floor(it hits the levator ani muscle and undergoes forward internal rotation through one- eighth of a circle to direct mento anterior position.
- The head born by flexion of the neck and head.
- in Mento transverse position (10-15%): **usually rotate anterior.**

-If the head enter the pelvis in mento- posterior position:

the head either rotate forward to change to mento-anterior position & deliver as mento anterior position.

- or the head undergoes backwards rotation to direct mento-posterior position.
- This position is **incompatible with vaginal delivery.**

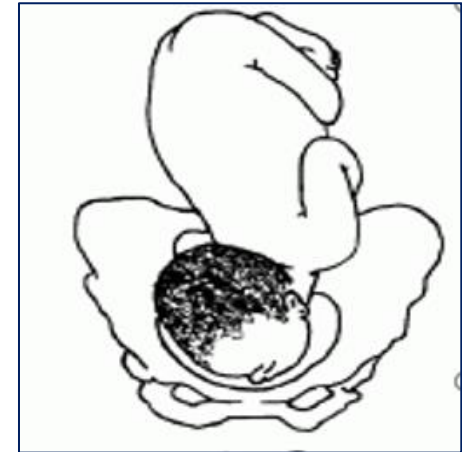
Diagnosis:



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Vaginal examination:

-The landmarks are: Mouth, jaws, nose, malar and orbital ridges.
presence of alveolar margins distinguishes
the mouth from the anus in breech presentation.



Management of labour:



The woman is kept in bed during the 1st stage.

- Once membranes ruptured, vaginal examination should be done to exclude cord prolapse.
- When the cervix becomes fully dilated, and the head reaches the pelvic floor, an episiotomy should be done especially in primigravida to avoid tearing of the perineum.

In mento-anterior and mento-transverse position:

- Spontaneous vaginal delivery is expected. If there is delay in the 2nd stage, deliver the baby by C/S.

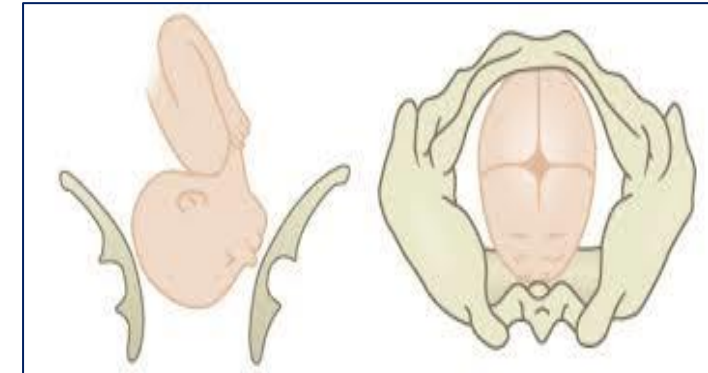
In mento-posterior position:

- Time should be allowed for spontaneous forward rotation to anterior position to take place (which happened in **45-65% of cases**).
- This usually happened late in the 2nd stage.
- If spontaneous rotation does not occur then **Caesarean section is the safest way of delivery to reduce fetal and maternal morbidity.**

Brow presentation (incidence 1/2000 deliveries):



- Brow presentation : when the head is partially extended
- Brow presentation is usually transient.
- With uterine contraction the head either undergoes more flexion and change to vertex presentation.
- Or further extension and change to face presentation.
- Persistent brow presentation is fortunately rare.
- In brow presentation the diameter which present to the pelvis is the mento-vertical which measure 13.5 cm.
- Such long diameter cannot engage and if persist obstructed labour would result.
- So there is no place for vaginal delivery in persistent brow presentation.



Diagnosis:



- Brow presentation should always be suspected when there is:

Non engagement of the fetal head in labour especially in woman who has had previous easy deliveries.

-As a rule there is early rupture of the membranes.

- On vaginal examination:

-The presenting part is usually high.

- The landmarks felt are:

1.The anterior fontanelle,

2.Frontal suture.

3.Forehead.

4.The orbital Ridges.

5.The bridge of the nose.

Management of labour:



- When brow diagnosed ante-natally by ultrasound and there is no evidence of disproportion or any other abnormality, nothing should be done as in most cases the head will flex during labour.
- If the head is discovered to be partially extended in early labour and there is no evidence of disproportion a short trial of labour is permitted, and this may result in further extension of the head to face presentation.
- If there is persistent brow presentation caesarean section should be done as vaginal delivery is impossible.

Compound presentation(1/1000 deliveries):



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Mean prolapse of fetal extremity alongside the presenting part.

It is three types:

- 1.Prolapse of the hand in cephalic presentation (the most common).**
- 2.Prolapse of upper extremity in breech presentation**
- 3.Prolapse of a lower extremity in cephalic presentation (is relatively rare).**

- Cord prolapse is common occurring in about 11-20% of cases of compound presentation.**
- The diagnosis usually made during labour by vaginal examination when the cervix dilated the prolapsed extremity can be felt alongside the head or breech.**
- Compound presentation should be suspected when there is poor progress in labour especially when the fetal head fails to engage during the active phase.**

Management:

This depend on:

1. Gestational age.
2. Type of presentation.



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- In cephalic presentation with prolapsed hand, labour can be allowed and vaginal delivery anticipated as the hand will move upward into the lower uterine segment as the vertex descend into the birth canal.
- Other compound presentation are treated by C/S.
- There is a risk of umbilical cord prolapse and continuous fetal heart monitoring should be done to detect fetal distress which should be treated by immediate caesarian section..

A lush green garden scene. In the foreground, a wooden bench with a dark metal frame sits on a grassy lawn. To the left of the bench is a large, dark rock. To the right is a small, round, dark-colored planter filled with white flowers. The background is filled with dense green foliage, including a large, leafy tree on the left and a bright yellow-green tree in the center. The sky is visible through the leaves at the top.

Thank you